

Module

3

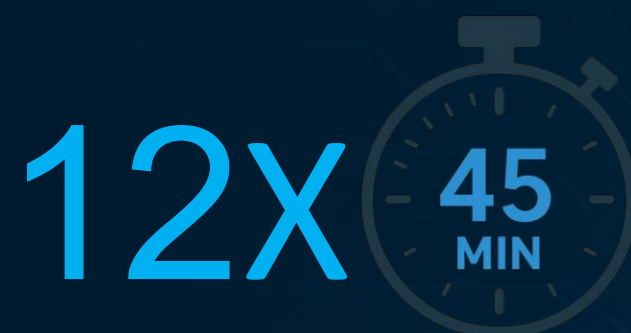
Operating Systems - Practical



Training structure



EASY



SIMULATION

Module: 3 - OS Fundamentals

- Module Statistics:
 - Total Hours: 25 (13 theory + 12 practical)
 - Number of Labs: 6 sessions
 - Operating Systems: Windows 11 & Linux (Kali)
 - Assessment: Hands-on labs + comprehensive security report

Module: 3 - OS Fundamentals

- Key Topics:
 - User and permission management
 - Security configuration and hardening
 - Event monitoring and log analysis
 - Security assessment and documentation

Lab Setup



Required VMs:

- Windows 11 VM
 - Administrator access required
 - Will be used for Sessions 1, 2, 5, 6
- Kali Linux VM
 - Root/sudo access required
 - Will be used for Sessions 3, 4, 5, 6
- Pre-installed Tools:
 - Wireshark (both VMs)
 - Cisco Packet Tracer (both VMs)
 - Standard OS tools and utilities
- Network Setup:
 - VMs on same virtual network
 - Must be able to ping each other

Safety and Best Practices 1

- Before Each Lab:
 - Take VM snapshots
 - Document starting configuration
 - Read entire lab instructions
 - Review related theory materials

Safety and Best Practices 2

- During Labs:
 - Follow instructions carefully
 - Take screenshots for documentation
 - Note errors or unexpected behavior
 - Ask for help if stuck >15 minutes

Safety and Best Practices 3

- After Each Lab:
 - Complete lab report
 - Take final snapshot
 - Review what you learned
 - Clean up (unless instructed otherwise)

Documentation Standards

- Cover page with details
- Executive summary (2-3 sentences)
- Step-by-step documentation
- Screenshots with annotations
- Answers to questions
- Reflection on learning

Windows Permissions



Session 1 - Objectives

- Learning Goals
 - Create users and groups
 - Configure NTFS permissions
 - Apply permission inheritance
 - Calculate effective permissions
 - Test access controls
 - Document configurations

Session 1 - Scenario

- Your Role: Junior System Administrator
 - Task: Configure department file shares
 - Requirements
- Three departments (HR, Finance, IT)
 - Department isolation enforced
 - Manager-only confidential folders
 - Shared folders for collaboration

User Creation

- Create 9 Users
 - HR: manager, staff1, staff2
 - Finance: manager, staff1, staff2
 - IT: manager, staff1, staff2
- Tool: Computer Management (compmgmt.msc)

Group Creation

- Create 6 Groups
 - HR_Department, HR_Managers
 - Finance_Department, Finance_Managers
 - IT_Department, IT_Managers
- Add Members Appropriately

Folder structure

- Create folder structure

Folder Structure

Create Directory Tree

```
C:\CompanyData\  
├── HR\  
│   ├── Shared\  
│   └── Confidential\  
├── Finance\  
└── IT\
```


Permission Configuration (1)

- Department Folders
 - Department Group: Read & Execute
 - Manager Group: Modify
 - Administrators: Full Control
 - Disable inheritance

Permission Configuration (2)

- Confidential Folders
 - Manager Group: Modify
 - Administrators: Full Control
 - Remove all other access
- Shared Folders
 - Department Group: Modify
 - Administrators: Full Control

Permission Testing

- Test Process
 - Switch user (Win+L)
 - Attempt file operations
 - Document results
 - Verify isolation
- Expected: Staff access Shared only, Managers access both

Effective Permissions

- What Are They?
 - Combined permission result
 - User + Group permissions
 - Most restrictive wins
- How to Check: Security → Advanced → Effective Access

Session 1 - Questions

- Question 1: Why use groups vs. direct permissions?
- Question 2: Difference between Deny and no permission?
- Question 3: What happens when moving files?
- Question 4: Why is inheritance useful?
- Question 5: How to grant temporary full access?

Session 1 - Questions

- Test Process
 - Switch user (Win+L)
 - Attempt file operations
 - Document results
 - Verify isolation
- Expected: Staff access Shared only, Managers access both

Session 1 - Deliverables

- User/group creation screenshots
- Folder structure diagram
- Permission configuration tables
- Testing results table
- Effective permissions screenshots
- Question answers

Session 1 - Summary

- Skills Gained
 - User and group management
 - NTFS permission configuration
 - Least privilege principle
 - Access control testing
 - Security documentation
- Next: Windows Security Configuration

Windows Security



Session 2 - Objectives

- Learning Goals
 - Configure Windows Firewall
 - Manage Windows Defender
 - Understand UAC
 - Enable security auditing
 - Analyze Event Viewer
 - Create security baseline

Session 2 - Scenario

- Your Role: Security Administrator
 - Task: Harden Windows 11 workstation
- Requirements
 - Block unnecessary services
 - Enable endpoint protection
 - Configure UAC
 - Implement monitoring

Windows Firewall (1)

- Inbound Rules
 - Block all by default
 - Allow RDP from specific IP
 - Allow File Sharing (local network)
 - Explicitly block Telnet (port 23)

Windows Firewall (2)

- Outbound Rules
 - Block suspicious IP ranges
 - Allow HTTP/HTTPS
 - Log dropped packets
- Tool: Windows Defender Firewall

Windows Defender

- Enable All Protection
 - Real-time protection
 - Cloud-delivered protection
 - Automatic sample submission
 - Tamper protection
- Test: Download EICAR test file

User Account Control

- Configure UAC
 - Set to "Always notify"
 - Test administrative tasks
 - Document user experience
- Levels: Always/Default/No dim/Never

Security Auditing (1)

- Enable Audit Policies
 - Account logon (Success & Failure)
 - Logon events (Success & Failure)
 - Object access (Success & Failure)
 - Policy change (Success)
 - Privilege use (Success & Failure)

Security Auditing (2)

- Configure File Auditing
 - Right-click Confidential folder
 - Security → Advanced → Auditing
 - Add "Everyone" principal
 - Select Full Control
 - Monitor all access

Event Viewer Analysis

- Key Event IDs
 - 4624: Successful logon
 - 4625: Failed logon
 - 4663: File access
 - 4688: Process creation
 - 4720: User created
 - 4732: Group membership change

Security Baseline

- Document Configuration
 - System information
 - Firewall rules
 - Defender settings
 - UAC configuration
 - Audit policies
 - Log retention

Session 2 - Questions

- Question 1: Inbound vs. outbound rules?
- Question 2: Why audit successes and failures?
- Question 3: Resolve Controlled Folder Access issues?
- Question 4: Real-time vs. cloud protection?
- Question 5: Event 4625 appears 50x in 5 min?

Session 2 - Deliverables

- Submit
 - Firewall configuration report
 - Defender settings documentation
 - UAC configuration
 - Audit policy screenshots
 - Sample events (4624, 4625, 4663)
 - Security baseline document

Session 2 - Summary

- Skills Gained
 - Host-based firewall rules
 - Endpoint protection management
 - Privilege elevation controls
 - Security event monitoring
 - Baseline documentation
- Next: Linux User & Permission Management

Linux Permissions



Session 3 - Objectives

- Learning Goals
 - Create users and groups
 - Understand rwx model
 - Use chmod and chown
 - Apply special permissions
 - Configure sudo access
 - Test access controls

Session 3 - Scenario

- Your Role: Linux System Administrator
 - Task: Multi-user environment setup
- Requirements
 - Department users and groups
 - Group-based access control
 - Shared project directories
 - Appropriate sudo privileges

User Creation

- Create 9 Users
 - sudo useradd -m -s /bin/bash username
 - sudo passwd username
- Verify: Check /etc/passwd and home directories

Group Management

- Create 6 Groups
 - sudo groupadd groupname
 - sudo usermod -aG groupname username
- Verify: groups username or id username

Linux Permission Model

- **rwx Permissions**
 - r (read): 4
 - w (write): 2
 - x (execute): 1
- Levels: Owner / Group / Others
- Example: `rwxr-xr--` = 754

Directory Setup

- Create Structure
 - `sudo mkdir -p /company/{hr,finance,it}/{shared,confidential}`
- Set Ownership
 - `sudo chown -R :hr_dept /company/hr`

Setting Permissions

- Shared Folders
 - `sudo chmod 770 /company/hr/shared`
- Confidential Folders
 - `sudo chmod 750 /company/hr/confidential`

Special Permissions

- SUID (4000): Execute as owner
- SGID (2000): Execute as group / Inherit group
- Sticky Bit (1000): Only owner deletes
- Apply SGID: `chmod 2770 directory`

Sudo Configuration

- Edit with visudo
 - sudo visudo
- Grant Access
 - Full sudo: %it_managers ALL=(ALL:ALL) ALL
 - Limited: %hr_managers ALL=/usr/sbin/useradd

Umask Configuration

- What is Umask?
 - Default permission mask
 - Subtracts from base (666 files, 777 dirs)
- Common Values
 - 022: Standard
 - 027: Restrictive
 - 077: Very restrictive

Permission Testing

- Test Scenarios
 - User accessing department folder
 - User accessing confidential
 - Cross-department access attempt
 - Manager sudo privileges
- Document results in matrix

Session 3 - Questions

- Question 1: Calculate: Base 777, umask 027
- Question 2: Purpose of SGID on directory?
- Question 3: Why sticky bit on /tmp?
- Question 4: Difference: chmod 755 vs. u+x?
- Question 5: Grant sudo for one command?

Session 3 - Deliverables

- Submit
 - User/group creation commands
 - Directory structure diagram
 - Permission configuration table
 - Special permissions implementation
 - Sudo configuration
 - Testing results

Session 3 - Summary

- Skills Gained
 - Linux user management
 - rwx permission model
 - chmod and chown usage
 - Special permissions (SUID/SGID/sticky)
 - Sudo configuration
 - Access control testing
- Next: Linux Security Hardening

Linux Hardening



Session 4 - Objectives

- Learning Goals
 - Secure SSH configuration
 - Configure UFW firewall
 - Manage system services
 - Implement fail2ban
 - Set up file integrity monitoring
 - Create security baseline

Session 4 - Scenario

- Your Role: Linux Security Administrator
 - Task: Harden Kali Linux system
- Requirements
 - Secure remote access
 - Host firewall configured
 - Unnecessary services disabled
 - Intrusion prevention active

SSH Hardening

- Edit Configuration
 - `sudo nano /etc/ssh/sshd_config`
- Key Changes
 - Port 2222 (non-default)
 - PermitRootLogin no
 - PasswordAuthentication no
 - MaxAuthTries 3
- Note: `man nano` for the handbook

SSH Key Setup

- Generate Key
 - `ssh-keygen -t ed25519`
 - `ssh-copy-id user@server`
- Restart SSH
 - `sudo systemctl restart ssh`

UFW Firewall

- Enable UFW
 - `sudo ufw enable`
 - `sudo ufw default deny incoming`
 - `sudo ufw default allow outgoing`

UFW Rules

- Allow Services
 - `sudo ufw allow 2222/tcp`
 - `sudo ufw allow 80/tcp`
 - `sudo ufw allow 443/tcp`
- Check Status: `sudo ufw status verbose`

Service Management

- List Services
 - `systemctl list-unit-files --type=service`
- Disable Unnecessary
 - `sudo systemctl stop service_name`
 - `sudo systemctl disable service_name`

Fail2Ban Setup

- Install
 - `sudo apt install fail2ban`
- Configure
 - `[sshd]`
 - `enabled = true`
 - `port = 2222`
 - `maxretry = 3`
 - `bantime = 3600`

Automatic Updates

- Install
 - `sudo apt install unattended-upgrades`
- Configure
 - Security updates: automatic
 - Other updates: manual

File Integrity (AIDE)

- Install & Initialize
 - `sudo apt install aide`
 - `sudo aideinit`
- Run Check
 - `sudo aide --check`

System Auditing

- Install auditd
 - `sudo apt install auditd`
- Example Rules
 - `--w /etc/passwd -p wa -k user_mod`
 - `--w /etc/sudoers -p wa -k sudoers`

Security Baseline

- Document
 - System information
 - SSH configuration
 - Firewall rules
 - Active services
 - Fail2ban settings
 - AIDE configuration

Session 4 - Questions

- Question 1: Why change default SSH port?
- Question 2: Purpose of Fail2Ban?
- Question 3: Stop vs. disable service?
- Question 4: Why file integrity monitoring?
- Question 5: Risks of automatic updates?

Session 4 - Deliverables

- Submit
 - SSH configuration
 - UFW rules documentation
 - Disabled services list
 - Fail2Ban configuration
 - AIDE scan results
 - Security baseline document

Session 4 - Summary

- Skills Gained
 - SSH security hardening
 - Host-based firewall (UFW)
 - Service management
 - Intrusion prevention (Fail2Ban)
 - File integrity monitoring (AIDE)
 - Security baseline creation
- Next: Event Log Analysis

Log Analysis



Session 5 - Objectives

- Learning Goals
 - Analyze Windows Event Viewer
 - Parse Linux system logs
 - Identify security anomalies
 - Correlate cross-platform events
 - Create analysis reports
 - Understand SIEM concepts

Session 5 - Scenario

- Your Role: SOC Analyst
- Investigation
 - Suspicious login attempts
 - Potential privilege escalation
 - Unusual file access
 - Cross-system correlation needed

Preparation

- On each of the system generate log events
 - Failed logins (note that there are multiple ways to do this. Experiment!)
 - Try to log in with explicit credentials
 - Try to log in with keys
 - Try to open files for which you have no rights
 - Sudo with an unknown user
- Make sure to access one system from the other
- Try to chain logins (from windows 2019 to windows 11, then from windows 11 to linux for example)
- Transfer files. Downloads, scp, share file transfers

Windows Event Analysis

- Key Event IDs
 - 4624: Successful logon
 - 4625: Failed logon
 - 4648: Explicit credential use
 - 4672: Special privileges
 - 4688: Process creation
 - 4720: User created

Failed Login Analysis

- Investigation Steps
 - Filter Event ID 4625
 - Analyze patterns
 - Identify source IPs
 - Count attempts per account
 - Check if attack succeeded

Linux Log Files

- Key Locations
 - /var/log/auth.log
 - /var/log/syslog
 - /var/log/kern.log
- List contents
 - less /var/log/auth.log
 - use the “q” command to quit, the “/” command to search
- Search Failed Logins
 - sudo grep "Failed password" /var/log/auth.log

Using journalctl

- Common Commands
 - `sudo journalctl -u ssh`
 - `sudo journalctl --since "1 hour ago"`
 - `sudo journalctl -f`
 - `sudo journalctl -p err`

Event Correlation

- Investigation Timeline format
 - Failed Windows login (10:15 AM UTC)
 - Successful Linux login (10:16 AM UTC)
 - File transfer observed (10:17 AM UTC)
- Match: Timestamps, usernames, IPs

Log Analysis Script

- Bash Example (try to explain the commands)

```
#!/bin/bash
```

```
echo "Failed attempts:"
```

```
sudo grep "Failed password" /var/log/auth.log | wc -l
```

```
echo "By IP:"
```

```
# ... parsing logic Investigate the use of "sort"
```

PowerShell Analysis

- Use the powershell editor. Get familiar. What is the difference between running as administrator and running as user for the following query events?
- Sample Query Events

```
Get-EventLog -LogName Security -InstanceId 4625
```

```
Get-EventLog -LogName Security | Group-Object EventID
```


Siem Concepts

- What is SIEM?
 - Centralized log collection
 - Real-time correlation
 - Automated alerting
 - Compliance reporting
- Examples: Splunk, ELK Stack, QRadar

Session 5 - Questions

- Question 1: Identify privilege escalation?
- Question 2: Event Viewer vs. journalctl?
- Question 3: Why log correlation?
- Question 4: Brute force attack patterns?
- Question 5: Log retention duration?

Session 5 - Deliverables

- Submit
 - Windows event analysis
 - Linux log analysis
 - Event correlation timeline
 - Log analysis scripts
 - Incident investigation summary
 - Key event screenshots

Session 5 - Summary

- Skills Gained
 - Windows Event Viewer analysis
 - Linux log parsing (auth.log, journalctl)
 - Log analysis script creation
 - Cross-platform correlation
 - SIEM concept understanding
- Next: System Hardening Assessment

Security Assessment



Session 6 - Objectives

- Learning Goals
 - Perform security assessments
 - Use automated scanning tools
 - Identify vulnerabilities
 - Prioritize remediation
 - Document findings
 - Create comprehensive reports

Session 6 - Scenario

- Your Role: Security Auditor
- Task: Comprehensive assessment
- Deliverable: Professional security report with findings, risks, and recommendations

Windows Assessment

- Manual Checks
 - Password policies (secpol.msc)
 - Installed software inventory
 - Running services review
 - Patch status verification
- PowerShell Audit
 - Get-HotFix
 - Get-Service | Where Status -eq "Running"

Linux Assessment (Lynis)

- Install & Run
 - `sudo apt install lynis`
 - `sudo lynis audit system`
- Review
 - Hardening index score
 - Suggestions by category
 - Warnings and findings

Vulnerability Documentation

- For Each Finding
 - Vulnerability name
 - System affected
 - Risk level (Critical/High/Medium/Low)
 - Description with evidence
 - Remediation recommendation

Risk Assessment

- Risk Matrix

	Low	Medium	High
High	Medium	High	Critical
Medium	Low	Medium	High
Low	Low	Low	Medium

- Factors: Likelihood × Impact

Compliance Checking

- Common Standards
 - CIS Benchmarks
 - NIST Cybersecurity Framework
 - ISO 27001
 - PCI DSS (if applicable)

Remediation Planning

- For Each Finding
 - Remediation steps
 - Resources required
 - Time estimate
 - Assigned owner
 - Verification method

Security Report

- Structure
 - Executive Summary
 - Scope & Methodology
 - Detailed Findings
 - Recommendations
 - Conclusion
- Professional formatting required. Actionable!

Remediation

- Use the remediation plan of your neighbor
- Apply to your own system.
 - Note what went wrong, what can be improved
 - Is the report complete?
 - Discuss!
- Do not remediate beyond what was in the remediation plan

Re-Assessment

- After Remediation and discussion
 - Re-run security scans
 - Verify fixes applied
 - Check for new issues
 - Update baseline

Section 6 - Questions

- Question 1: Why not rely only on automated tools?
- Question 2: Prioritize with limited resources?
- Question 3: Vulnerability vs. risk?
- Question 4: Why document after remediation?
- Question 5: Assessment frequency?

Session 6 - Deliverables

- Submit
 - Windows assessment report
 - detailed review on the security assessment of your neighbor. Pros, cons and recommendations
 - Linux assessment (Lynis)
 - Consolidated vulnerability list
 - Risk assessment matrix
 - Remediation action plan
 - Final baselines (both systems)
 - Executive summary

Session 6 - Summary

- Skills Gained
 - Security assessment methodology
 - Automated scanning tools
 - Vulnerability identification
 - Risk assessment techniques
 - Remediation planning
 - Professional security reporting

Wrap Up



Module 4 Complete!

- Sessions Completed
 - Windows Permissions
 - Windows Security
 - Linux Permissions
 - Linux Hardening
 - Log Analysis
 - Security Assessment
- Core Skills: Cross-platform administration, security hardening, monitoring, assessment

Final Assessment

- Grading
 - 6 lab reports (50%)
 - Final security report (30%)
 - Report on remediation (10%)
 - Questions & reflections (10%)
- Criteria: Technical accuracy, documentation quality, professionalism

Congratulations!

- You've Mastered
 - Windows & Linux security fundamentals
 - Access control implementation
 - Security monitoring & analysis
 - Professional documentation
- Next Steps: Apply to Module 4, build labs, pursue certifications

Thank you